

Unit 4B

1. Describe the each of the indicated characteristics of the function $f(x) = 4 + 6 \sin(3x + \pi)$.

a) Transformations: Up 4, left $\frac{\pi}{3}$, Horiz. Comp., Vert. Stretch

b) Period: $\frac{2\pi}{3}$

c) The y-intercept is at (0, 4)

d) Minimum y-value is 10 and maximum y-value is -2
 $4+6=10$ $4-6=-2$

$$f(x) = 4 + 6 \sin\left(3\left(x + \frac{\pi}{3}\right)\right)$$

$$\text{Period} = \frac{2\pi}{B} = \frac{2\pi}{3}$$

$$f(0) = 4 + 6 \sin(\pi) = 4 + 6(0)$$

2. Evaluate each inverse trig expression.

a) $\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right) = \frac{3\pi}{4}$

b) $\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$

c) $\tan^{-1}(-\sqrt{3}) = -\frac{\pi}{3}$

$\theta \in \text{Q I, II}$

$\theta \in \text{Q I, IV}$

$\theta \in \text{Q I, IV}$

3. Evaluate each composite expression.

a) $\sin^{-1}(\sin(\frac{3\pi}{2})) = -\frac{\pi}{2}$

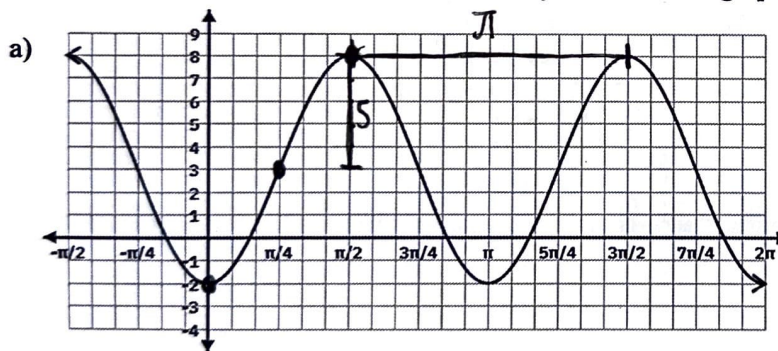
b) $\cos^{-1}(\cos(\frac{5\pi}{6})) = \frac{5\pi}{6}$ $\cos^{-1}(-\frac{\sqrt{3}}{2})$

c) $\cos^{-1}(\sin(\frac{4\pi}{3})) = \frac{5\pi}{6}$

d) $\sin^{-1}(\cos(\frac{11\pi}{6})) = \frac{\pi}{3}$ $\sin^{-1}(\frac{\sqrt{3}}{2})$

e) $\tan^{-1}(\cos(\pi)) = -\frac{\pi}{4}$

4. Write a sinusoidal function that accurately describes the graph given below. You may use either sine or cosine.



Starts at min, use -cosine

$$Y_{\min} = -2$$

$$\text{Amplitude} = 5$$

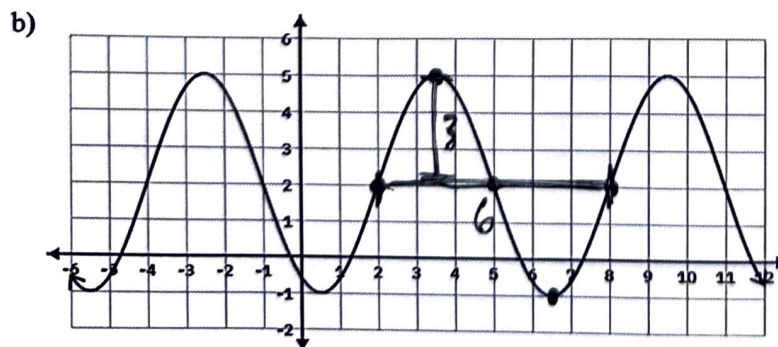
$$Y_{\max} = 8$$

$$\text{Period} = \pi \Rightarrow B = \frac{2\pi}{P} = \frac{2\pi}{\pi} = 2$$

$$Y_{\text{middle}} = 3$$

No phase shift

$$f(x) = 3 - 5 \cos(2x)$$



Use +sine Shifted right by 2 units

$$Y_{\min} = -1$$

$$\text{Amplitude} = 3$$

$$Y_{\max} = 5$$

$$\text{Period} = 6 \Rightarrow B = \frac{2\pi}{P} = \frac{\pi}{3}$$

$$Y_{\text{middle}} = 2$$

$$f(x) = 2 + 3 \sin\left(\frac{\pi}{3}(x-2)\right)$$

* Answers vary depending on what function and phase shift you choose.